Rayleigh-Taylor Instability using a VOF-Style Method

~23BME013

Aim: To study the growth of a perturbed heavy-over-light interface while monitoring phase-fraction boundedness and conservation.

Case definition and parameters

Simulations were analyzed on the following parameters.

Parameter	Value
Coordinate system	Cartesian
Grid points, nx	96
Grid points, ny	192
nu	0.0008
T_end	2
Atwood number	0.2, 0.5, 0.8
Gravity	1
Poisson_iteration	60

Governing Equation:

The transported phase indicator α satisfies

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \mathbf{u}) = 0,$$

while the Boussinesq momentum equation contains a buoyancy term proportional to the Atwood number

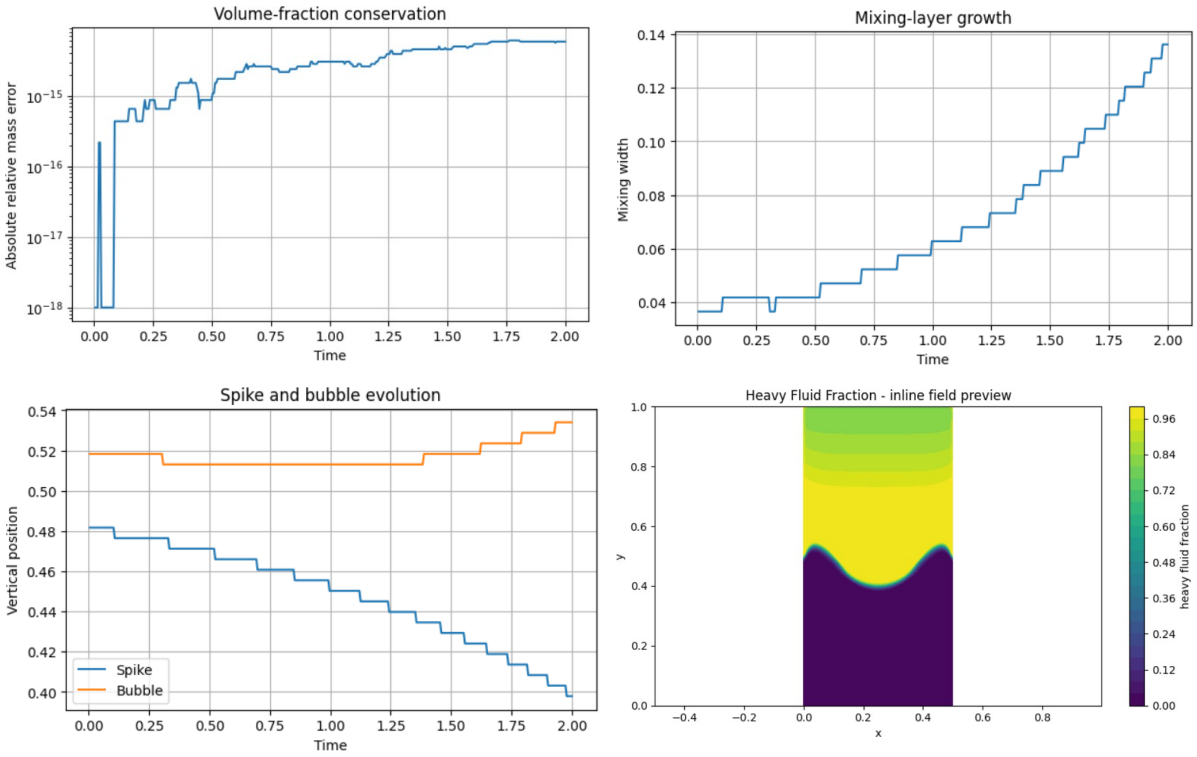
$$A = \frac{\rho_h - \rho_l}{\rho_h + \rho_l}.$$

Boundary Conditions: No-slip impermeable box boundaries; initially perturbed interface with heavier fluid above lighter fluid.

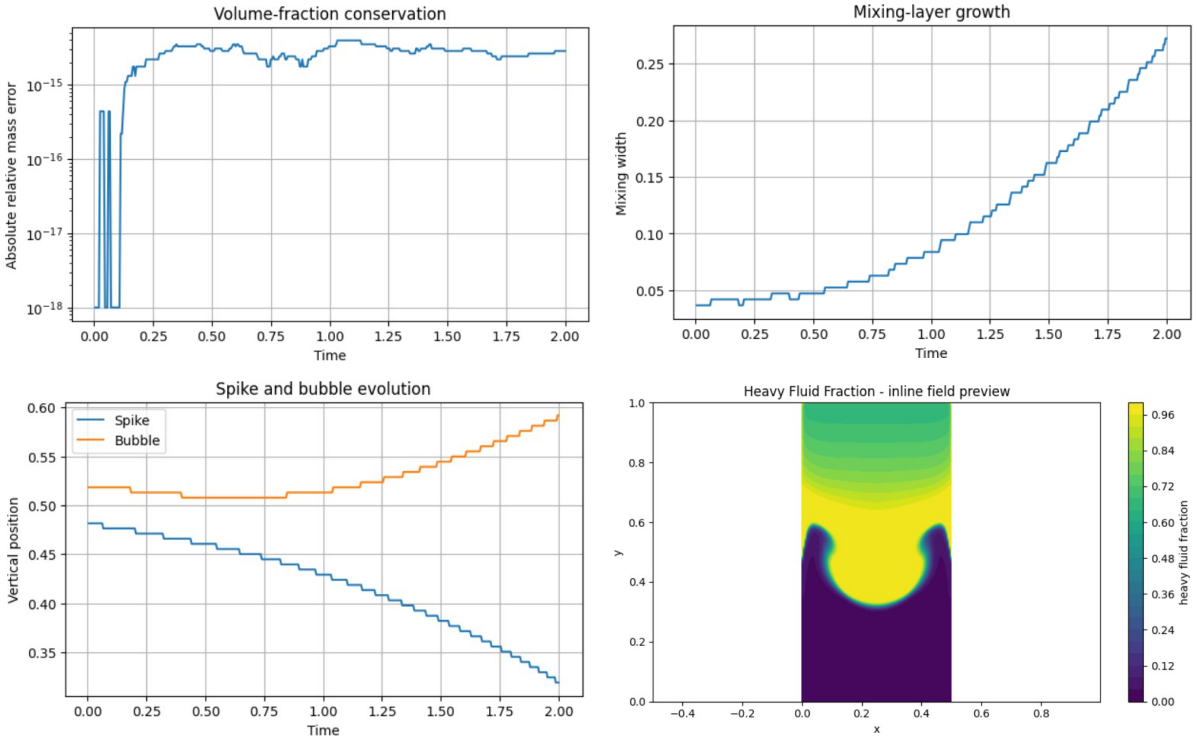
Numerical Method: Projection flow solver coupled to conservative upwind phase-fraction transport, clipping, and conservative redistribution to maintain $0 \leq \alpha \leq 1$.

Observations: Spike and bubble positions, mixing width, divergence, mass error, minimum and maximum phase fraction, and interface shape at Atwood numbers 0.2, 0.5, and 0.8. Compare spike, bubble, mixing width, boundedness, and conservation.

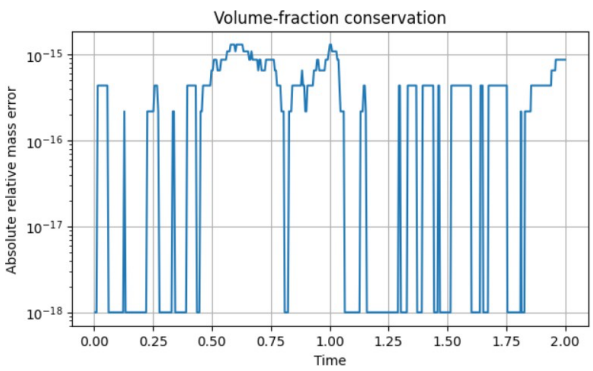
For Atwood number 0.2:

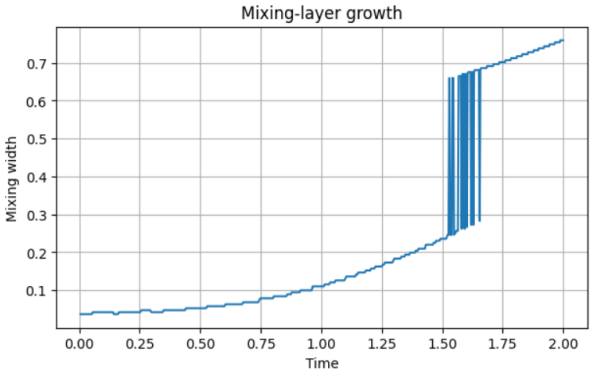


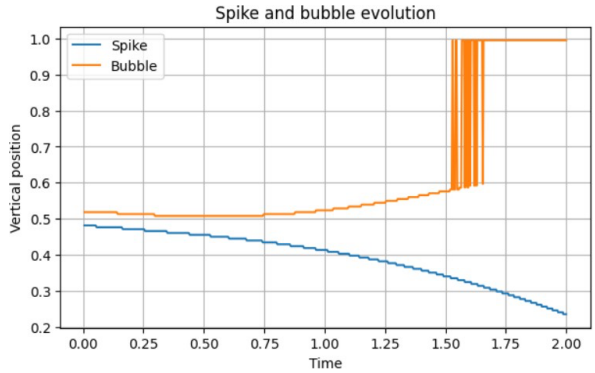
For Atwood number 0.5:

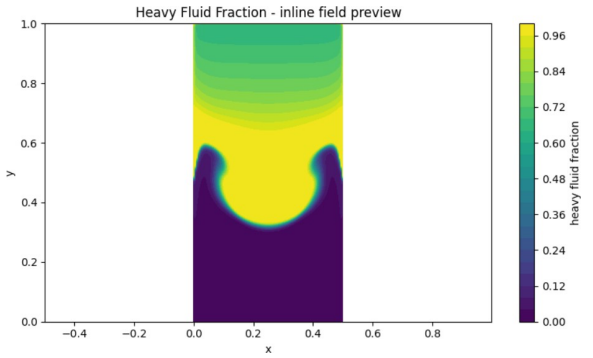


For Atwood number 0.8:









Results:

Parameter	**A = 0.2**	**A = 0.5**	**A = 0.8**
Interface at t = 0 (spike / bubble position, y)	≈ 0.48 / 0.52	≈ 0.48 / 0.52	≈ 0.48 / 0.52
Spike position at t = 2 (y)	≈ 0.398	≈ 0.319	≈ 0.235
Spike descent from t = 0 to 2	≈ 0.084	≈ 0.163	≈ 0.247
Bubble position at t = 2 (y)	≈ 0.534	≈ 0.592	≈ 0.58 at t ≈ 1.5, then reads 1.0 (artefact)
Mixing width at t = 0	≈ 0.037	≈ 0.037	≈ 0.037
Mixing width at t = 2	≈ 0.136	≈ 0.272	≈ 0.76 (affected by artefact)
Net mixing-width growth	≈ 0.099 (at t = 2)	≈ 0.235 (at t = 2)	≈ 0.20 (at t = 1.5)
Maximum absolute relative mass error	≈ 6×10^{-15}	≈ 4×10^{-15}	≈ 1.3×10^{-15}
Jumps in bubble position / mixing width	none	none	t ≈ 1.5–1.65
Phase-fraction range, $0 \leq \alpha \leq 1$ (field plot)	≈ 0 to 1	≈ 0 to 1	[INSERT: min/max α]
Heavy-fluid fraction near top wall at t = 2	≈ 0.8	≈ 0.6	[INSERT: from field plot]
Interface shape at t = 2	smooth, single spike at mid-width	rounded spike, curled tips at side walls	[INSERT: from field plot]

Conclusion:

The Rayleigh–Taylor instability of a single-mode perturbed heavy-over-light interface in a no-slip box was simulated using a projection flow solver with conservative upwind phase-fraction transport, at Atwood numbers of 0.2, 0.5 and 0.8. In all cases the heavy fluid above penetrates downward as a spike at mid-width, while the lighter fluid rises along the side walls as bubbles. The interface starts at $y \approx 0.5$ with the spike at $y \approx 0.48$ and the bubble at $y \approx 0.52$. By $t = 2$ the spike has descended to $y \approx 0.398$, 0.319 and 0.235 for $A = 0.2$, 0.5 and 0.8 respectively, i.e. by about 0.084, 0.163 and 0.247. The bubble has risen to $y \approx 0.534$ for $A = 0.2$ and $y \approx 0.592$ for $A = 0.5$.

Effect of Atwood number on growth: Because the buoyancy term scales with the Atwood number, spike descent, bubble rise and mixing width all increase with A. The spike descent grows less than in proportion to A, in the ratio of about 1 : 1.9 : 2.9 for A in the ratio 1 : 2.5 : 4. The mixing width, which equals the bubble position minus the spike position in the plotted curves, grows from ≈ 0.037 at $t = 0$ to ≈ 0.136 ($A = 0.2$) and ≈ 0.272 ($A = 0.5$) at $t = 2$. The net growth of ≈ 0.099 and ≈ 0.235 is in the ratio of about 2.4, close to the Atwood-number ratio of 2.5. Growth is slow at first and accelerates: for $A = 0.2$ the net growth at $t = 2$ is about 3.8 times that at $t = 1$, close to quadratic in time, while for $A = 0.5$ the ratio is about 5. In every case the spike moves faster than the bubble, and the bubble lags most at $A = 0.2$. The bubble position first decreases slightly, by about 0.005–0.01, before it rises, which is consistent with the initial smoothing of the interface. The stair-step appearance of the curves reflects the discrete grid cells to which the positions are tied.

Interface shape: At $A = 0.2$ the interface remains smooth and close to a single-mode shape, with the spike at mid-width and the bubbles at the side walls. At $A = 0.5$ the spike has a rounded head, and the rising light fluid at the side walls develops curled tips, which indicates the onset of the nonlinear roll-up stage. This stage appears earlier and more strongly as the Atwood number increases.

Conservation and boundedness: The absolute relative mass error of the volume fraction stays at round-off level, below approximately 6×10^{-15}, 4×10^{-15} and 1.3×10^{-15} for $A = 0.2$, 0.5 and 0.8, because both the transport and the redistribution are conservative. The phase fraction stays within $0 \leq \alpha \leq 1$ in the field plots. However, conservation of the total phase fraction does not guarantee a physical field. In the field plots the heavy-fluid fraction in the upper layer decreases progressively towards the top wall, to about 0.8 for $A = 0.2$ and about 0.6 for $A = 0.5$ at $t = 2$, although no interface is present there. For an incompressible flow with an exactly divergence-free velocity this region should remain at $\alpha = 1$. This points to a residual velocity divergence acting through the conservative transport term, and/or to the clipping and redistribution step spreading phase fraction, so the boundedness is partly enforced by redistribution and not obtained from accurate transport alone.

Result at A = 0.8: The spike continues to descend to $y \approx 0.235$, but between $t \approx 1.5$ and 1.65 the bubble position repeatedly jumps from ≈ 0.6 to the top wall ($y = 1$), and the mixing width jumps from ≈ 0.27 to ≈ 0.7. After this the bubble remains at $y = 1$ and the mixing width reads ≈ 0.76 at $t = 2$, which equals $1.0 - 0.235$. Since the heavy-fluid fraction near the top wall is depleted more strongly as A increases, these jumps are most likely caused by the interface-detection criterion picking up the diluted top layer, not by physical bubble rise. The spike, bubble and mixing-width values for $A = 0.8$ are therefore reliable only up to about $t = 1.5$.

Overall, the simulations reproduce the expected trend that Rayleigh–Taylor growth strengthens with the Atwood number, with a nearly proportional increase of the net mixing-width growth between $A = 0.2$ and 0.5, and conserve the phase fraction to round-off at every Atwood number. The depletion of the heavy-fluid fraction near the top wall and the artefact in the $A = 0.8$ interface metrics limit the quantitative reliability of the results. Checking the velocity divergence, defining the spike and bubble positions from the $\alpha = 0.5$ contour near the interface instead of from the global extent of the mixed region, and refining the grid would improve the quantitative reliability.